✅ STEP 0: Open Git Bash

```
cd ~
```

✅ STEP 1: Create project folder

```
mkdir data-science-project
cd data-science-project
```

✅ STEP 2: Initialize Git repo

```
git init
```

✓ Creates local Git repository

✅ STEP 3: Create first file (program 1)

```
touch prog1.py
notepad prog1.py
```

✍️ Code inside prog1.py:

```
num = int(input("Enter number: "))
for i in range(1, 11):
    print(num, "x", i, "=", num*i)
```

✅ STEP 4: Stage + commit

```
git add .
git commit -m "Multiplication Table program"
```

✅ STEP 5: Connect GitHub repo

Go to github and create a repo data-science-project and then copy the path:

```
git remote add origin https://github.com/your-username/data_science_project.git
git remote -v
```

✅ STEP 6: Push to GitHub (master → remote)

```
git push -u origin master
```

⚠️ OPTIONAL (modern standard)

Rename master → main:

```
git branch -M main
git push -u origin main
```

✅ STEP 7: Create new branch (GIT-EXAM)

```
git checkout -b GIT-EXAM
git branch
```

✅ STEP 8: Create second program

```
touch prog2.py
notepad prog2.py
```

✍️ Code inside prog2.py:

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

if a > b:
  print("Greatest:", a)
else:
  print("Greatest:", b)
```

✅ STEP 9: Add + commit branch changes

```
git add .
git commit -m "Greatest of two numbers program"
```

✅ STEP 10: Push branch to GitHub

```
git push -u origin GIT-EXAM
```

✅ STEP 11: Switch back to master/main

```
git checkout master
```

(or)

```
git checkout main
```

✅ Correct diff command

```
git diff master GIT-EXAM
```

✓ This shows:

- prog2.py exists only in GIT-EXAM
- master does NOT have it

✅ STEP 12: View full history

```
git log --oneline --graph --all
```